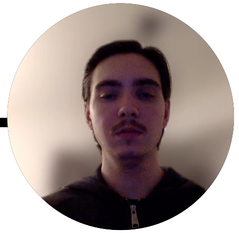

## TABLEAU ASSIGNMENT:

# THE IMPACT OF COVID-19 PANDEMIC ON ROAD SAFETY


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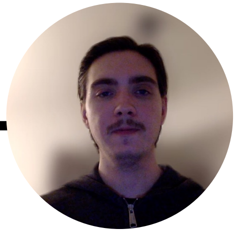


# MOTIVATION

### Understanding COVID-19's impact on road safety in Finland

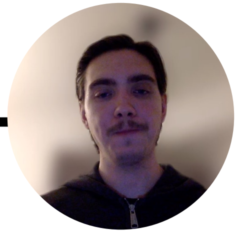
- The pandemic significantly altered daily life, including how and when people travel.
- It's unclear whether these changes led to safer or more dangerous roads.
- A data-driven analysis is crucial to separate real trends from assumptions.

### Targeting long-term traffic safety improvements

- Identifying accident patterns during the pandemic can reveal hidden risks or protective factors.
- These insights can inform permanent safety measures beyond COVID-19.
- Helps shift focus from reactive to proactive road safety strategies.

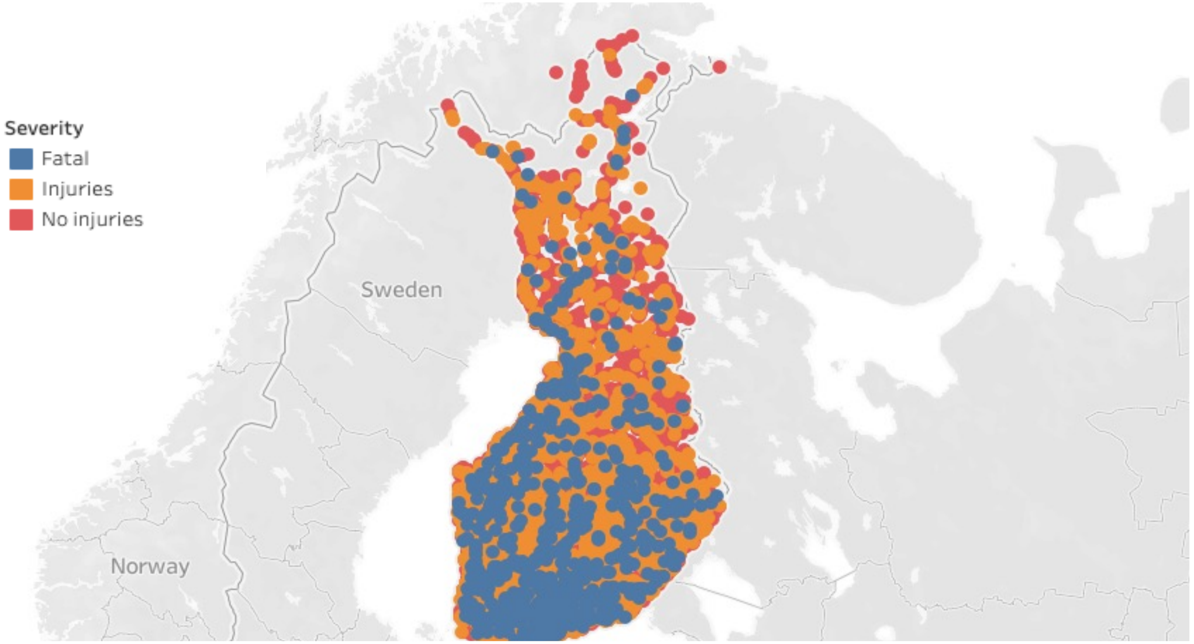
### Supporting evidence-based policy and planning

- Authorities need reliable data to make informed decisions about infrastructure and traffic policies.
- Data-driven findings can support smarter urban mobility planning and emergency readiness.
- Ensures future strategies are guided by trends, not just temporary observations.

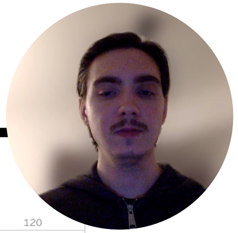


# GEOGRAPHIC DISTRIBUTION OF TRAFFIC ACCIDENTS

Finnish highways



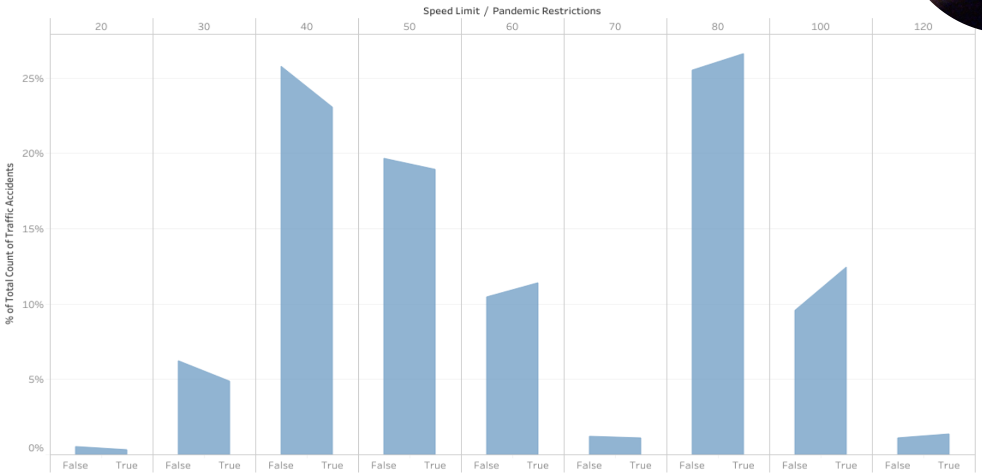

- The most serious accidents are concentrated along highways.
- In sparsely populated areas, the lack of traffic supervision likely plays a major role.
- In the north, road conditions may be a particularly significant factor.



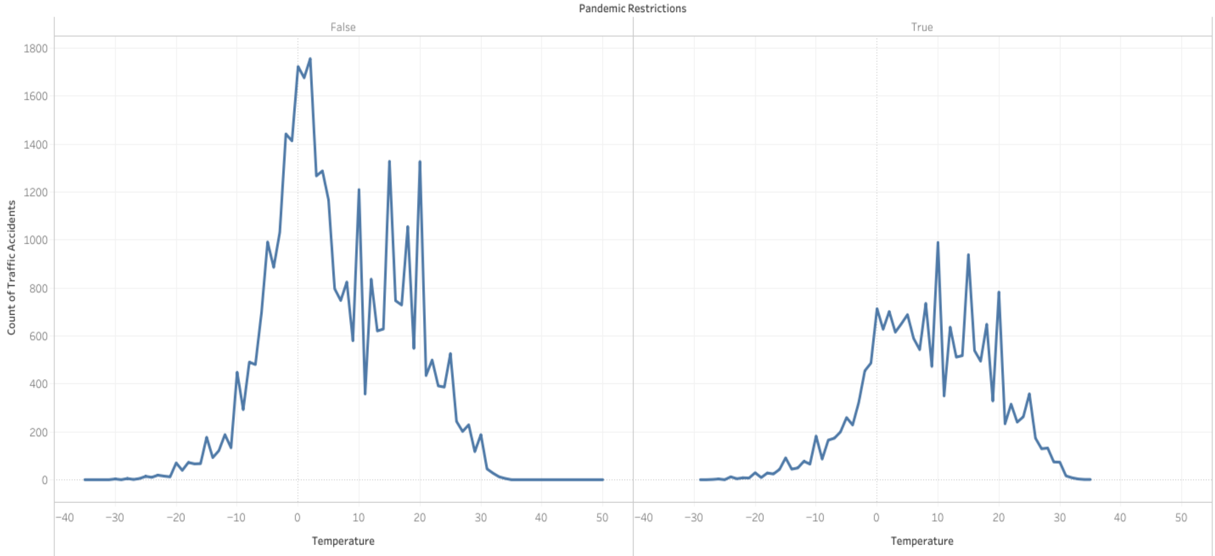
# SHIFTING PATTERNS IN ACCIDENT PREVALENCE DURING THE PANDEMIC

Pandemic restrictions disrupted the seasonal pattern of temperature-related accidents

Distribution of Traffic Accidents by Speed Limit by Pandemic Restrictions

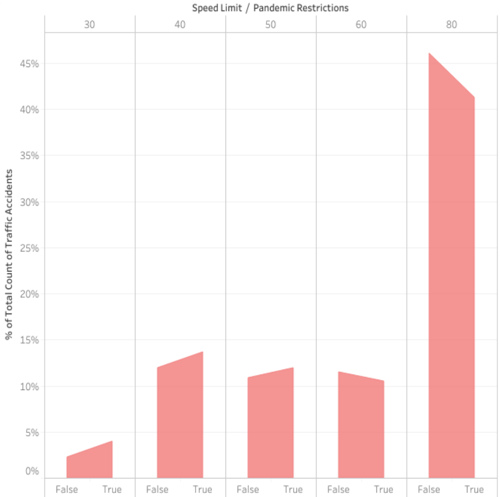



Traffic Accidents Seasonality




Distribution of Lethal Traffic Accidents by Speed Limit by Pandemic Restrictions

*



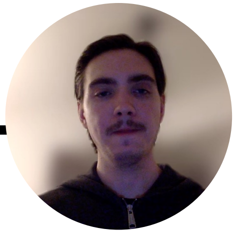

The relative prevalence of accidents shifts around the 50 km/h speed limit, indicating a pivot in distribution trends (5,08 % prevalence distribution shift from instances less or equal to 50 km/h)

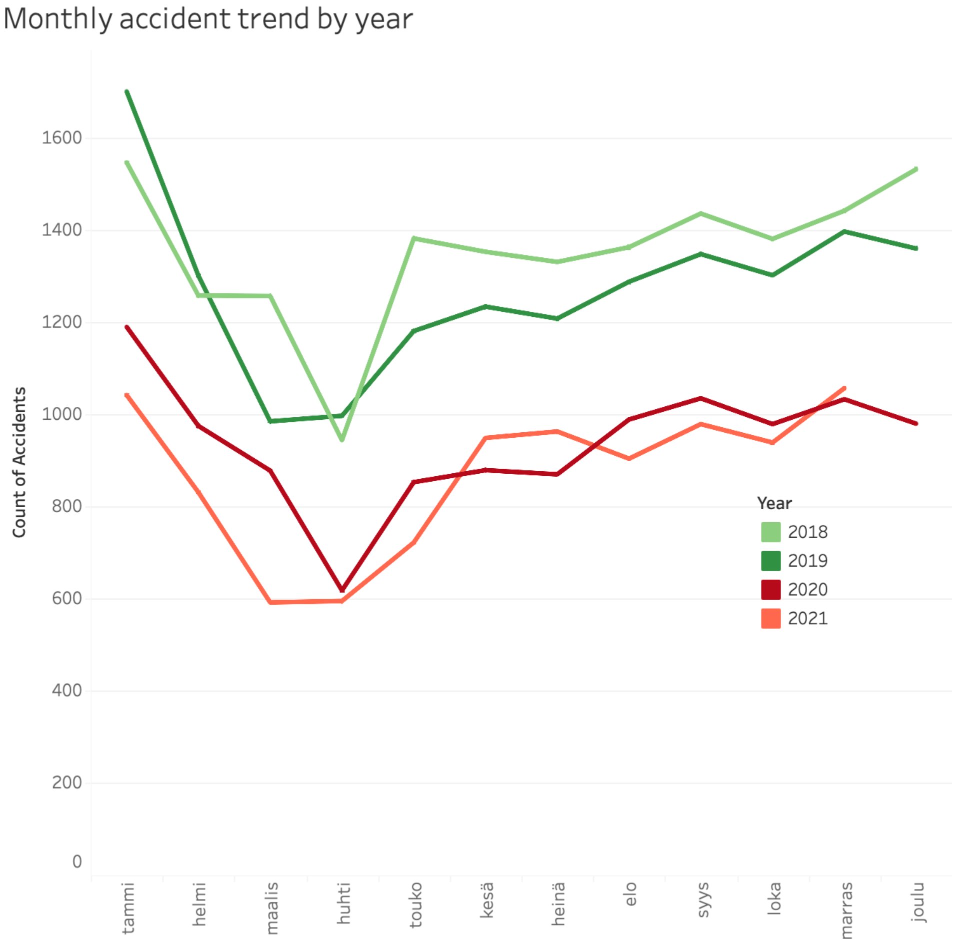
- <--However, the distribution trend for fatal accidents follows an inverse relationship (4,58 % prevalence distribution shift from instances greater or equal to 60 km/h)

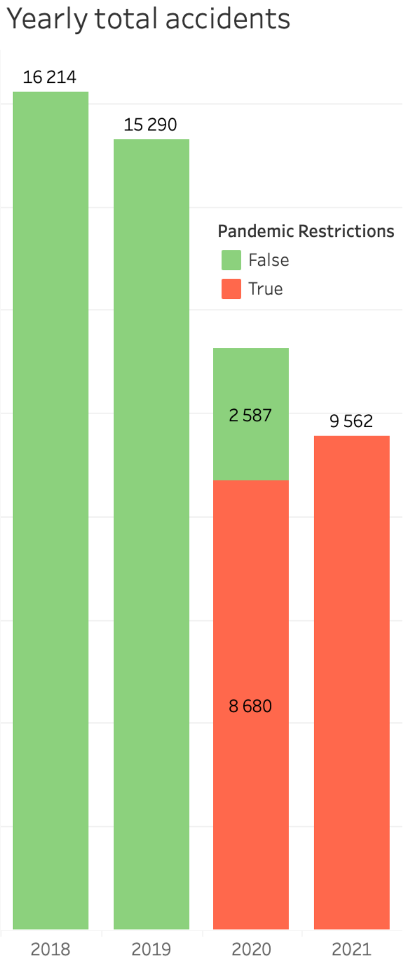
*Excluding speed limits 70,100,120 from visualization for greater clarity

# ACCIDENT TREND DEVELOPMENT

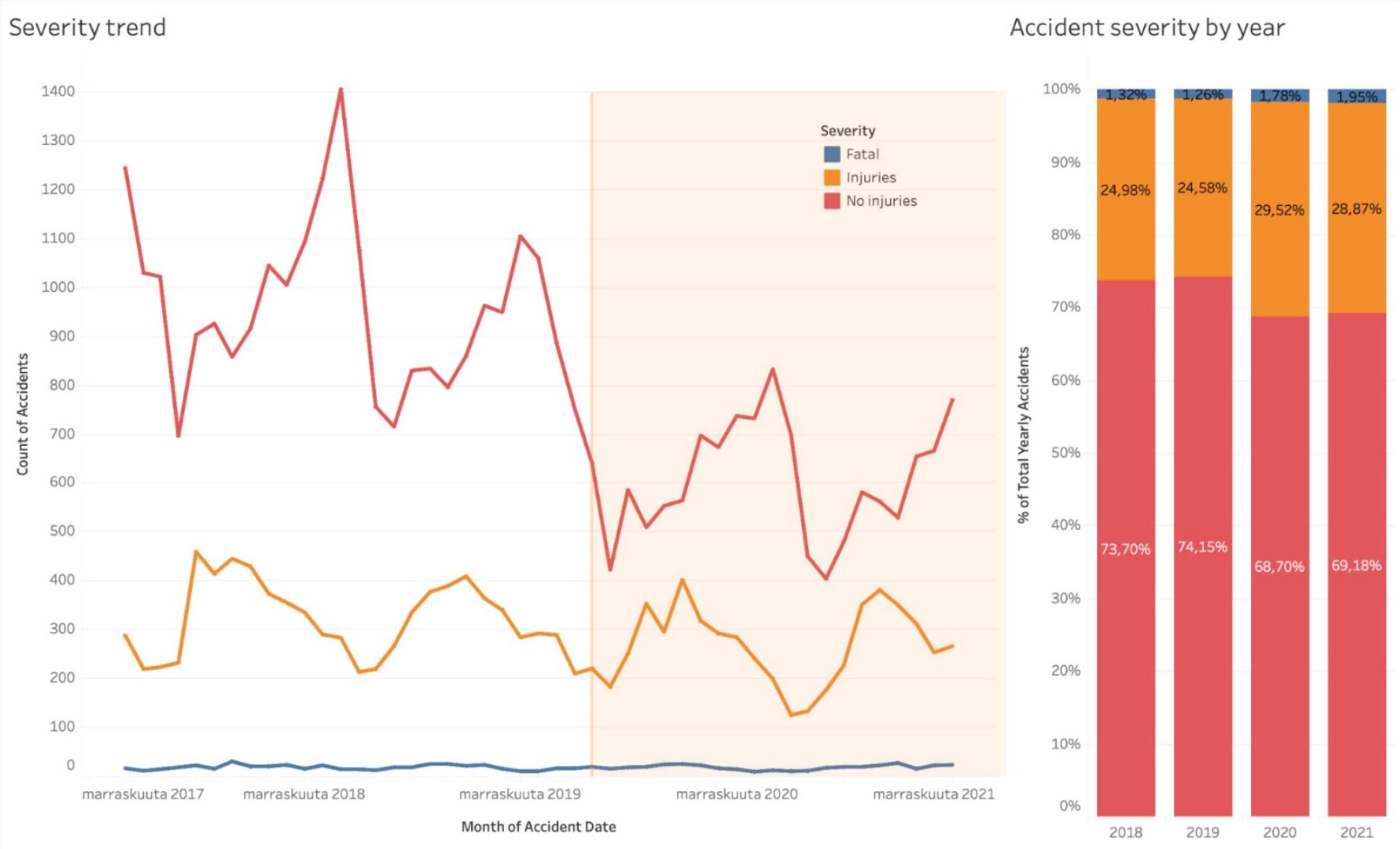
- **Yearly traffic accidents decreased by 40% (over 6000 yearly accidents) from 2018 to 2021 and by 37% from 2019 to 2021**
    - Some of the decrease may be explained by a general decreasing trend in traffic accidents
- All years show a seasonal trend.
- In 2020, a clear drop occurred in March–April, likely due to the onset of the COVID-19 pandemic.
- Accidents were consistently highest in 2019, especially in January and February.
- 2021 had the lowest monthly accident counts overall, remaining below all previous years throughout.





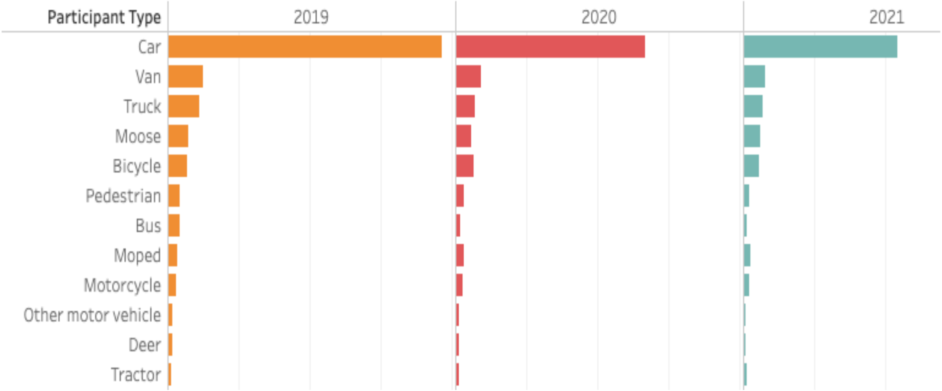
# CHANGE IN ACCIDENT SEVERITY OVER THE PANDEMIC



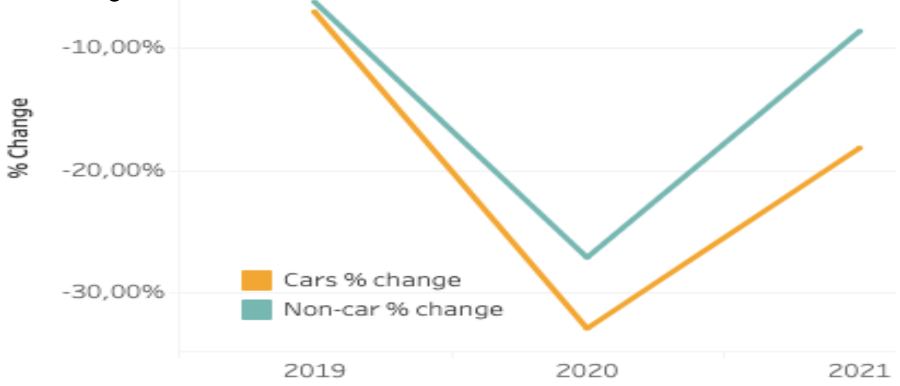

- Although the general number of accidents significantly decreased during the pandemic**, the relative proportion of severe accidents increased**
- From 2019 to 2020 **the relative proportion of fatal accidents increased by 55% and accidents leading to injuries by 18%**
- The relative proportion of accidents leading to no injuries decreased by 8,82% from 2019 to 2020
    - Note: The decreasing trend began already in January before pandemic restrictions held

# CHANGE IN ACCIDENT PARTICIPANT TYPE OVER THE PANDEMIC

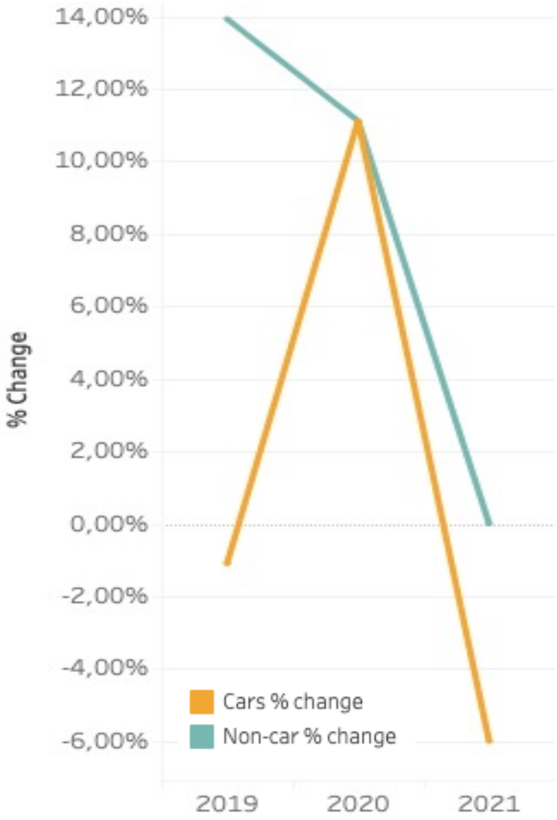
Participant types in all accidents




% Change of all accidents




% Change of fatal accidents




**Fatal accidents increased more steeply for cars** than for non-car participants in 2020, despite an overall drop in traffic volumes.

This may reflect **riskier driving behavior** in emptier roads, such as speeding or impaired driving during the pandemic.

In 2021, fatal car accidents dropped significantly, while non-car fatalities remained relatively stable

*Non-car includes all participant types except cars

# CHANGES IN ACCIDENT CATEGORIES DURING COVID RESTRICTIONS

- Moose and bicycle accidents became two of the six most common accident categories during COVID-19 restrictions, compared to before restrictions

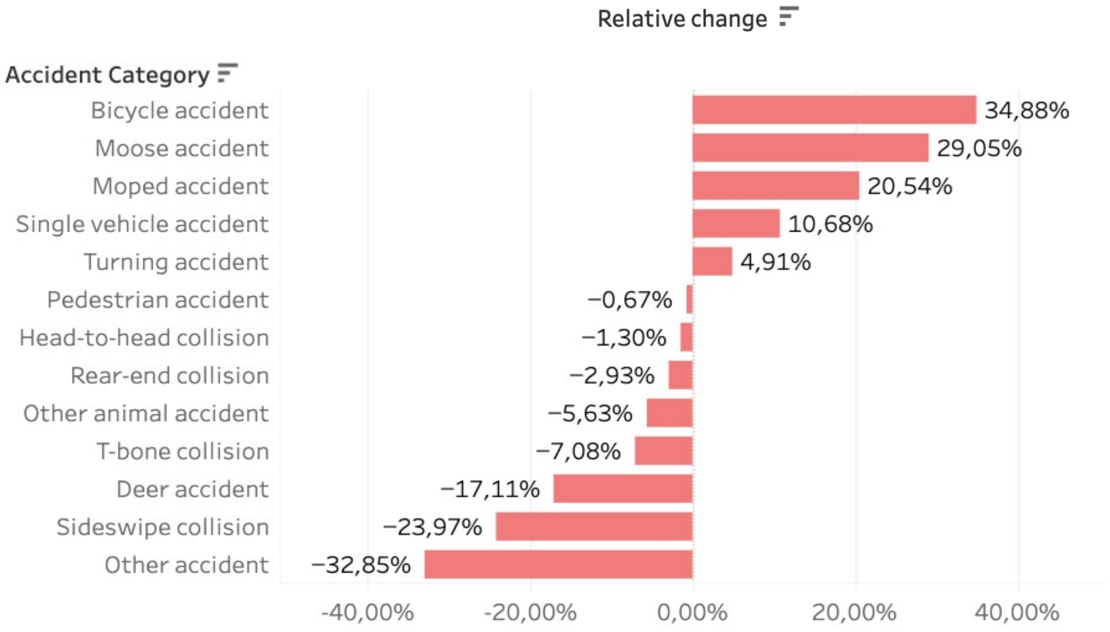
- Bicycle, pedestrian, and head-to-head collisions are the three most deadly* accident categories, making the rise in bicycle accidents a potential contributor to the increase in severe accidents

Pandemic Restrictions: True

| Accident Category | |
|---|---|
| Single vehicle accident | 26,95% |
| Moose accident | 13,15% |
| Other accident | 10,71% |
| Rear-end collision | 10,26% |
| T-bone collision | 7,87% |
| Bicycle accident | 6,96% |

Pandemic Restrictions: False

| Accident Category | |
|---|---|
| Single vehicle accident | 24,35% |
| Other accident | 15,95% |
| Rear-end collision | 10,57% |
| Moose accident | 10,19% |
| T-bone collision | 8,47% |
| Sideswipe collision | 7,01% |



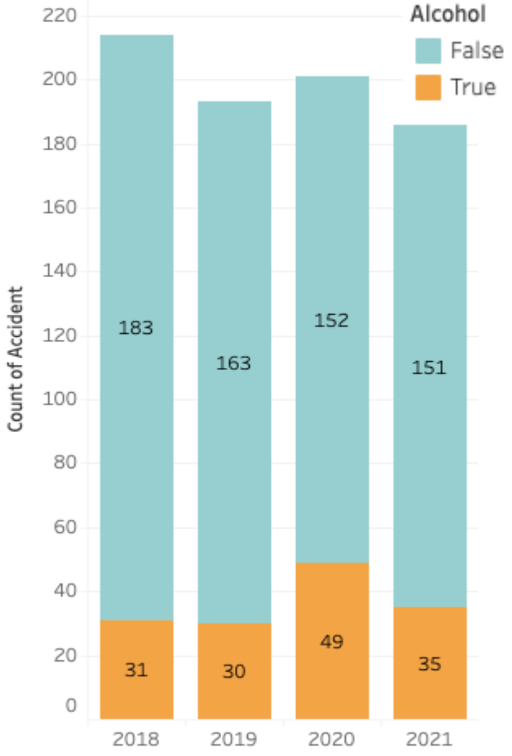

In terms of relative change** during pandemic restrictions:

- The share of bicycle, moose, and moped accidents saw a clear increase
- The share of single-vehicle accidents increased by 10%
- The share of deer accidents, sideswipes, and other accident types experienced a significant decrease during the restriction period

*calculated from all the fatal accidents in the data
**calculation: (accident category % during restrictions / accident category % outside of restrictions ) -100%

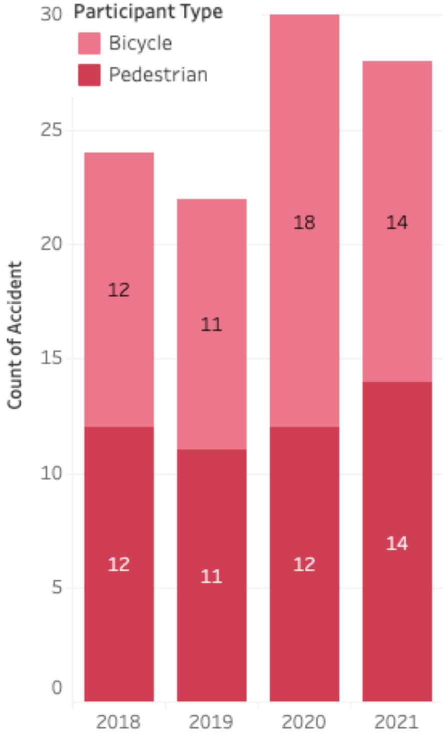
# FATAL ACCIDENTS DURING COVID-19: KEY CONTRIBUTING FACTORS

Alcohol-Related and Other Fatal Accidents




Fatal Alcohol-Related Accidents by Hour and Year


Fatal Accidents Involving Cyclists and Pedestrians by Year



# KEY INSIGHTS

### Accident Trends & Severity

**Data:**
Total accidents dropped by **~40%** (approximately 6000 fewer) from 2018–2021, yet fatal accidents increased propotionally by **54.8%** and severe (non-fatal) collisions by **18%** from 2019 to 2020, while non-severe accidents declined by **8.82%**.

**Insight:**
Fewer collisions occurred overall, but the **remaining crashes were significantly more severe**—suggesting that reduced traffic volumes may have inadvertently encouraged riskier driving behaviors.

### Shifts in Participant & Accident Types

**Data:**
Car accidents fell sharply (a **32.92%** reduction from 2019 to 2020), while non-car incidents involving cyclists, pedestrians, and mopeds either **declined less or increased proportionally**; additionally, incidents like moose and bicycle collisions became more frequent during COVID-19 restrictions.

**Insight:**
Pandemic-related lifestyle changes (e.g., increased outdoor activity, avoidance of public transport) shifted the road user mix toward more **vulnerable groups** and altered accident types, contributing to **higher injury severity**.

### Risky Driving Behaviors & Alcohol

**Data:**
Emptier roads during the pandemic saw an increase in high-speed and **risk-prone driving**, while remote work and changes in social routines led to a notable shift in **alcohol consumption in nightime and weekends**.

**Insight:**
These behavioral shifts—combined with less congested roads—resulted in a **disproportionate rise in severe and fatal accidents**, even as overall collision counts dropped.

# LONG-TERM RECOMMENDATIONS:
# HOW TO REDUCE THE NUMBER OF ACCIDENTS OVERALL

- **Target high-risk time periods**:

Focus enforcement and awareness campaigns on weekends and nighttime, when fatal and alcohol-related accidents are more prevalent.

- **Road-Specific Enforcement & Infrastructure Improvements:**

Use targeted enforcement (e.g., digital speed monitoring) on low traffic & rural roads to maximize limited resources and consider basic infrastructure upgrades on high-risk roads and highways.

- **Strengthen Protection for Vulnerable Road Users:**

Improve infrastructure, control speeds, and promote safety awareness for drivers and high-risk collision participants, such as pedestrians, cyclists, and other non-car travelers.

- **Monitor & Reinforce Safe Driving Behaviors:**

Regularly analyze traffic data and coordinate among agencies to adapt policies swiftly, while launching education campaigns that foster responsible driving habits long after COVID.